17

Technical data

Contents of this chapter

This chapter contains the technical specifications of the drive, for example, the ratings, sizes and technical requirements, provisions for fulfilling the requirements for CE and other markings.

Ratings

The nominal rating for the drive modules with 50 Hz and 60 Hz supply are given below.

IEC RATINGS										
Drive type ACS880-34-	**Frame size**	**Input current**	**Output ratings**							
			Nominal use				**Light-duty use**		**Heavy-duty use**	
		I_1	I_{max}	I_2	P_N	S_n	I_{Ld}	P_{Ld}	I_{Hd}	P_{Hd}
		A	**A**	**A**	**kW**	**kVA**	**A**	**kW**	**A**	**kW**
U_N = 400 V										
246A-3	R11	212	350	246	132	170	234	132	206	110
293A-3	R11	257	418	293	160	203	278	160	246	132
363A-3	R11	321	498	363	200	251	345	200	293	160
442A-3	R11	401	545	442	250	306	420	250	363	200
505A-3	R11	401	560	505	250	350	480	250	363	200
585A-3	R11	505	730	585	315	405	556	315	442	250
650A-3	R11	569	730	650	355	450	618	355	505	250
U_N = 500 V										
240A-5	R11	169	350	240	132	208	228	132	180	110
260A-5	R11	205	418	260	160	225	247	160	240	132

IEC RATINGS										
Drive type ACS880-34-	**Frame size**	**Input current**	**Output ratings**							
			Nominal use				**Light-duty use**		**Heavy-duty use**	
		I_1	I_{max}	I_2	P_N	S_n	I_{Ld}	P_{Ld}	I_{Hd}	P_{Hd}
		A	**A**	**A**	**kW**	**kVA**	**A**	**kW**	**A**	**kW**
361A-5	R11	257	542	361	200	313	343	200	260	160
414A-5	R11	321	542	414	250	359	393	250	361	200
460A-5	R11	404	560	460	315	398	450	315	414	250
503A-5	R11	455	560	503	355	436	492	355	460	315
U_N = 690 V										
142A-7	R11	123	250	123	132	170	135	132	119	110
174A-7	R11	149	274	149	160	208	165	160	142	132
210A-7	R11	186	384	186	200	251	200	200	174	160
271A-7	R11	232	411	232	250	324	257	250	210	200
330A-7	R11	293	480	293	315	394	320	315	271	250
370A-7	R11	330	520	330	355	442	360	355	330	315
430A-7	R11	375	520	375	400	514	420	400	370	355
										3AXD00000588487

UL (NEC) RATINGS								
Drive type ACS880-34-	**Frame size**	**Input current**	**Max. current**	**Output ratings**				
				App. power	**Light-duty use**		**Heavy-duty use**	
		I_1	I_{max}	S_n	I_{Ld}	P_{Ld}	I_{Hd}	P_{Hd}
		A	**A**	**kVA**	**A**	**hp**	**A**	**hp**
U_N = 480 V								
240A-5	R11	169	350	208	240	200	180	150
260A-5	R11	205	418	225	260	200	240	200
302A-5	R11	239	498	262	302	250	260	200
361A-5	R11	257	542	313	361	300	302	250
414A-5	R11	321	542	359	414	350	361	300
460A-5	R11	404	560	398	430	350	414	350
503A-5	R11	455	560	436	483	400	483	400
U_N = 575 V								
142A-7	R11	125	250	170	144	150	125	125
174A-7	R11	149	274	208	168	175	144	150
210A-7	R11	186	384	251	200	200	174	175
271A-7	R11	232	411	324	257	250	210	200
330A-7	R11	293	480	394	320	300	271	250
370A-7	R11	330	520	442	360	350	330	300
430A-7	R11	375	520	514	420	450	370	350
								3AXD00000588487

U_N	Nominal voltage of the drive
I_1	Nominal input current (rms) at 40 °C (104 °F)
S_n	Apparent power (no overload)
I_{max}	Maximum output current. Available for 10 seconds at start, otherwise as long as allowed by drive temperature. 140% … 200% of I_{Hd}, depending on power rating.
I_2	Continuous rms output current. No overload capability at 40 °C (104 °F). This is indicated in the type designation label as output current I_2.
P_N	Typical motor power in no-overload use
I_{Ld}	Continuous rms output current allowing 10% overload for 1 minute every 5 minutes
P_{Ld}	Typical motor power for light-duty use
I_{Hd}	Continuous rms output current allowing 50% overload for 1 minute every 5 minutes
P_{Hd}	Typical motor power for heavy-duty use
Note: To achieve the rated motor power given in the table, the rated current of the drive must be higher than or equal to the rated motor current. The power ratings apply to most IEC 34 motors at the nominal voltage of the drive. ABB recommends to select the drive, motor and gear combination for the required motion profile with the DriveSize dimensioning tool available from ABB.	

When is derating necessary

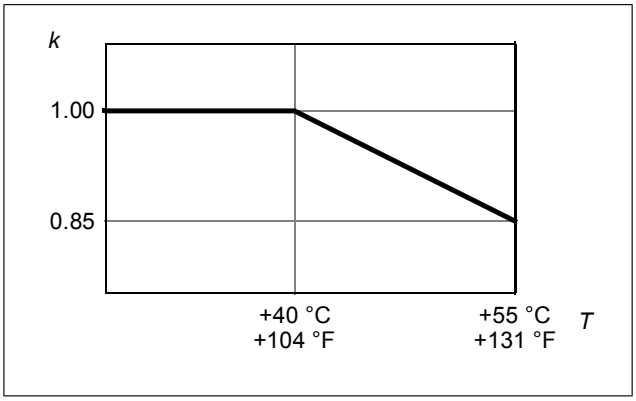
Derate the continuous output current of the drive if

- ambient temperature exceeds +40 °C (+104 °F) or
- drive is installed higher than 1000 m (3280 ft) above sea level
- switching frequency is other than default
- the minimum requirements of motor cable length are not met (see chapter d*u*/d*t* filters and sine filters).

Note: The final derating factor is a multiplication of all applicable derating factors.

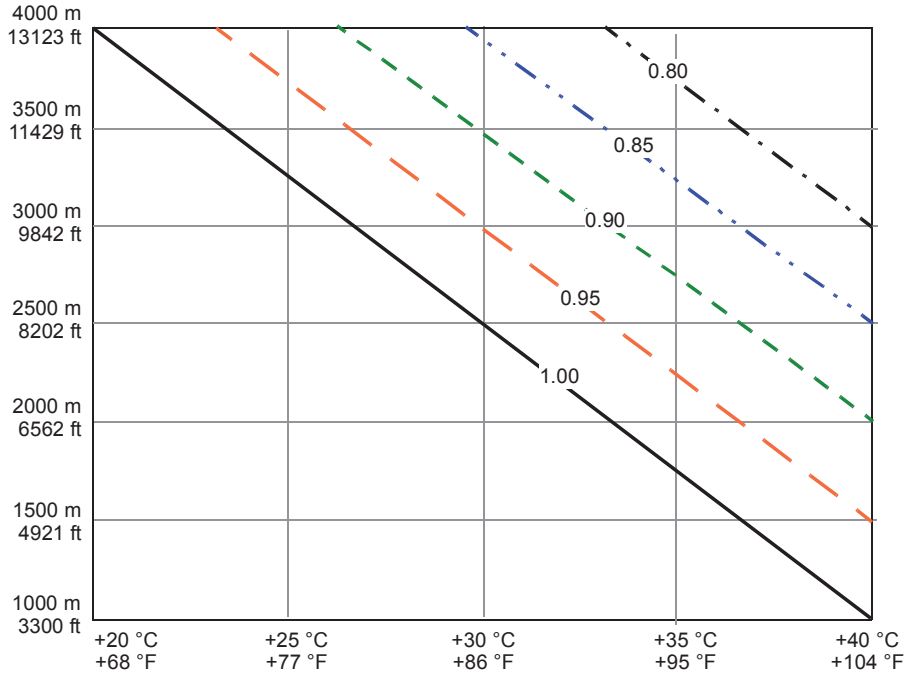
Ambient temperature derating

In the temperature range +40…55 °C (+104…131 °F), the rated output current is derated by 1% for every added 1 °C (1.8 °F) as follows. Calculate the output current by multiplying the current given in the rating table by the derating factor.



Altitude derating

At altitudes from 1000 to 4000 m (3300 to 13123 ft) above sea level, the derating is 1% for every 100 m (328 ft). If ambient temperature is below +40 °C (+104 °F), the derating can be reduced by 1.5% for every 1 °C reduction in temperature. For a more accurate derating, use the DriveSize PC tool. A few altitude derating curves are shown below.



	Derating factor 1.00
	Derating factor 0.95
	Derating factor 0.90
	Derating factor 0.85
	Derating factor 0.80

Deratings for special settings in the drive control program

Enabling special settings in the drive control program can require output current derating.

Ex motor, sine filter, low noise

Table below gives the deratings for these cases:

- drive is used with an ABB motor for explosive atmospheres (Ex) and **EX motor** in parameter **95.15 Special HW settings** is enabled
- sine filter given in the selection table (see section *Sine filters*) is used and **ABB sine filter** in parameter **95.15 Special HW settings** is enabled
- **Low noise optimization** is selected in parameter **97.09 Switching freq mode**.

With other than recommended sine filters (see section *Sine filters*) and non-ABB Ex motors, contact ABB.

Drive type ACS880-34-	Output ratings for special settings											
	Ex motor (ABB Ex motor)				ABB sine filter				Low noise mode			
	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use
	I_N	P_N	I_{Ld}	I_{Hd}	I_N	P_N	I_{Ld}	I_{Hd}	I_N	P_N	I_{Ld}	I_{Hd}
	A	kW	A	A	A	kW	A	A	A	kW	A	A
U_N = 400 V												
246A-3	234	132	222	196	221	132	210	185	217	132	204	180

Drive type ACS880-34-	Output ratings for special settings											
	Ex motor (ABB Ex motor)				ABB sine filter				Low noise mode			
	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use
	I_N	P_N	I_{Ld}	I_{Hd}	I_N	P_N	I_{Ld}	I_{Hd}	I_N	P_N	I_{Ld}	I_{Hd}
	A	kW	A	A	A	kW	A	A	A	kW	A	A
293A-3	278	160	264	237	264	160	251	221	258	160	243	215
363A-3	345	200	328	278	327	200	310	264	320	200	301	256
442A-3	420	250	399	345	398	250	378	327	390	250	367	317
505A-3	480	315	456	345	455	250	432	327	445	250	419	317
585A-3	556	315	528	420	527	315	500	398	516	315	485	386
650A-3	618	355	587	480	585	355	556	455	573	315	539	441
U_N = 460 V												
302A-5	287	250 hp	287	247	272	250 hp	272	234	266	250 hp	264	277
U_N = 500 V												
240A-5	228	132	217	171	216	132	205	162	212	132	199	157
260A-5	247	160	235	228	234	160	222	216	229	160	216	210
361A-5	343	200	326	247	235	200	309	234	318	200	300	227
414A-5	393	250	373	343	373	250	354	325	365	250	343	315
460A-5	437	315	428	393	414	315	405	373	406	250	393	362
503A-5	478	355	467	437	453	315	443	414	443	315	430	402
U_N = 690 V												
142A-7	125	132	119	105	128	132	122	107	66	75	63	55
174A-7	153	160	145	125	157	160	149	128	81	90	77	66
210A-7	185	200	176	153	189	200	180	157	98	110	93	81
271A-7	238	250	226	185	244	250	231	189	126	132	119	98
330A-7	290	315	282	238	297	315	288	244	154	160	149	126
370A-7	326	355	317	290	333	355	324	297	172	200	167	153
430A-7	378	400	370	326	387	400	378	333	200	200	195	172
												3AXD00000588487

U_N	Nominal voltage of the drive
I_N	Continuous rms output current. No overload capability at 40 °C (104 °F)
P_N	Typical motor power in no-overload use
I_{Ld}	Continuous rms output current allowing 10% overload for 1 minute every 5 minutes
P_{Ld}	Typical motor power for light-duty use
I_{Hd}	Continuous rms output current allowing 50% overload for 1 minute every 5 minutes

High speed mode

The selection **High speed mode** of parameter **95.15 Special HW settings** improves control performance at high output frequencies. ABB recommends it to be selected with output frequency of 120 Hz and above.

This table gives the drive module ratings for 120 Hz output frequency and the maximum output frequency for the drive ratings when **High speed mode** in parameter **95.15 Special HW settings** is enabled: With output frequencies smaller than this recommended maximum output frequency, the current derating is less than the values given in the table. Contact ABB for operation above the recommended maximum output frequency or for the output current derating with output frequencies above 120 Hz and below the maximum output frequency.

Drive type ACS880-34-	Deratings with selection High speed mode of parameter 95.15 Special HW settings									
	120 Hz output frequency					Maximum output frequency				
	Frequency	Nominal use		Light-duty use	Heavy-duty use	Maximum frequency	Nominal use		Light-duty use	Heavy-duty use
	f	I_N	P_N	I_{Ld}	I_{Hd}	f_{max}	I_N	P_N	I_{Ld}	I_{Hd}
	Hz	A	kW	A	A	Hz	A	kW	A	A
U_N = 400 V										
246A-3	120	246	132	234	206	500	201	110	193	170
293A-3	120	293	160	278	246	500	240	132	229	203
363A-3	120	363	200	345	293	500	297	200	284	241
442A-3	120	442	250	420	363	500	362	250	346	299
505A-3	120	505	250	480	363	500	413	250	395	299
582A-3	120	585	315	556	442	500	479	315	458	364
650A-3	120	650	355	618	505	500	532	315	509	416
U_N = 460 V										
302-5	120	302	250 (hp)	302	260	500	247	200 (hp)	249	214
U_N = 500 V										
240A-5	120	240	132	228	180	500	196	132	188	148
260A-5	120	260	160	247	240	500	213	160	203	198
361A-5	120	361	200	343	260	500	295	250	283	214
414A-5	120	414	250	393	361	500	339	250	324	297
460A-5	120	460	315	450	414	500	376	315	371	341
503A-5	120	503	355	492	460	500	412	315	405	379
U_N = 690 V										
142A-7	120	142	132	135	119	500	82	75	78	68
174A-7	120	174	160	165	142	500	100	110	95	82
210A-7	120	201	200	200	174	500	121	132	115	100
271A-7	120	271	250	257	210	500	156	160	148	121
330A-7	120	330	315	320	271	500	190	200	184	156
370A-7	120	370	355	360	330	500	213	250	207	190
430A-7	120	430	400	420	370	500	247	250	241	213
										3AXD00000588487

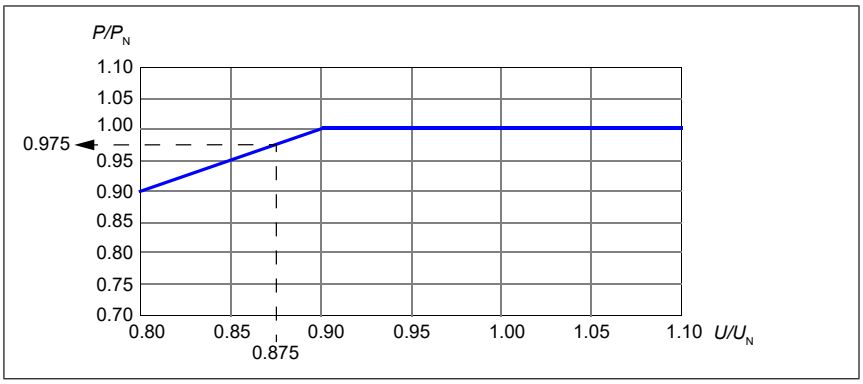
f	Output frequency
f_{max}	Maximum output frequency with High speed mode
U_N	Nominal voltage of the drive

I_N	Continuous rms output current. No overload capability at 40 °C (104 °F).
P_N	Typical motor power in no-overload use
I_{Ld}	Continuous rms output current allowing 10% overload for 1 minute every 5 minutes
I_{Hd}	Continuous rms output current allowing 50% overload for 1 minute every 5 minutes

Derating for output voltage boosting

The drive can output a higher motor voltage than the supply voltage. This can require derating of the drive output power depending on the difference between the supply voltage and the output voltage to the motor for continuous operation.

This drawing shows the required derating for -3 and -5 (400 V and 500 V) drive types.



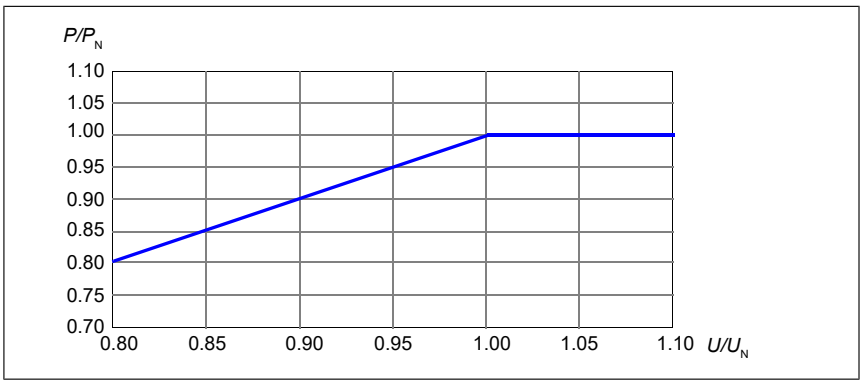
Example 1: P_N for ACS880-34-650A-3 is 355 kW. The input voltage (U) is 350 V. -> U/U_N = 350 V / 400 V = 0.875. -> P/P_N = 0.975 -> The derated power P = 0.975 × 355 kW = 346 kW.

To boost the output voltage to correspond the nominal supply voltage 400 V, increase the DC voltage to 400 V × √2 × 1.04 = 588 V.

Example 2: P_N for ACS880-34-503A-5 is 355 kW. The input voltage (U) is 450 V. -> U/U_N = 450 V / 500 V = 0.9. -> P/P_N = 1.00 -> The derated power P = 1.00 × 355 kW = 355 kW.

To boost the output voltage to correspond the nominal supply voltage 500 V, increase the DC voltage to 500 V × √2 × 1.04 = 735 V.

This drawing shows the required derating for -7 (690 V and 575 V) drive types.



To boost the output voltage to correspond the nominal supply voltage 690 V, increase the DC voltage to 690 V × √2 × 1.04 = 1015 V.

U	Input voltage of the drive

U_N	Nominal supply voltage of the drive. For -3 types U_N = 400 V, for -5 types U_N = 500 V. For -7 types U_N = 690 V but 575 V when P_N refers to nominal power ratings in the UL (NEC) 575 V rating table.
P	Derated ouput power of the drive
P_N	Nominal power rating of the drive

Fuses (IEC)

aR fuses by Cooper Bussmann for protection against short-circuit in the input power cable of drive are listed below.

Ultrarapid (aR) fuses						
Drive type ACS880-34-	**Input current (A)**	**Fuse**				
		A	**A²s**	**V**	**Type DIN 43653**	**Size**
U_N = 400 V						
246A-3	212	400	74000	690	170M5408	2
293A-3	257	500	145000	690	170M5410	2
363A-3	321	630	210000	690	170M6410	3
442A-3	401	700	300000	690	170M6411	3
505A-3	401	800	465000	690	170M6412	3
585A-3	505	1000	945000	690	170M6414	3
650A-3	569	1000	945000	690	170M6414	3
U_N = 500 V						
240A-5	169	315	42000	690	170M4410	1
260A-5	205	400	74000	690	170M5408	2
302A-5	249	500	145000	690	170M5410	2
361A-5	257	630	210000	690	170M6410	3
414A-5	321	700	300000	690	170M6411	3
460A-5	404	700	300000	690	170M6411	3
503A-5	455	800	465000	690	170M6412	3
U_N = 690 V						
142A-7	123	250	21000	690	170M4409	1
174A-7	149	315	42000	690	170M4410	1
210A-7	186	400	74000	690	170M5408	2
271A-7	232	500	145000	690	170M5410	2
330A-7	293	630	210000	690	170M6410	3
370A-7	330	700	300000	690	170M6411	3
430A-7	375	700	300000	690	170M6411	3
3AXD00000588487						

Note:

- See also section *Protecting the drive and input cable in short-circuits and against thermal overload (page 86)*.
- In multicable installations, install only one fuse per phase (not one fuse per conductor).
- Fuses with higher current rating than the recommended ones must not be used. Fuses with lower current rating can be used.
- Fuses from other manufacturers can be used if they agree with the ratings and the melting curve of the fuse does not exceed the melting curve of the fuse mentioned in the table.

Fuses (UL Recognized)

UL Recognized fuses for branch circuit protection per NEC are listed below. **Check that the operating time of the fuse is below 0.1 seconds.** The operating time depends on the fuse type, supply network impedance and the cross-sectional area, material and length of the supply cable. The fuses must be of the “non-time delay” type. Obey local regulations.

Drive type ACS880-34-	Input current (A)	Fuse				
		A	V	Manufacturer	Size	Type DIN 43653
U_N = 400 V						
246A-3	212	400	690	Bussmann	2	170M5408
293A-3	257	500	690	Bussmann	2	170M5410
363A-3	321	630	690	Bussmann	3	170M6410
442A-3	401	700	690	Bussmann	3	170M6411
505A-3	401	800	690	Bussmann	3	170M6412
585A-3	505	1000	690	Bussmann	3	170M6414
650A-3	569	1000	690	Bussmann	3	170M6414
U_N = 500 V						
240A-5	169	315	690	Bussmann	1	170M4410
260A-5	205	400	690	Bussmann	2	170M5408
302A-5	249	500	690	Bussmann	2	170M5410
361A-5	257	630	690	Bussmann	3	170M6410
414A-5	321	700	690	Bussmann	3	170M6411
460A-5	404	700	690	Bussmann	3	170M6411
503A-5	455	800	690	Bussmann	3	170M6412
U_N = 690 V						
142A-7	123	250	690	Bussmann	1	170M4409
174A-7	149	315	690	Bussmann	1	170M4410
210A-7	186	400	690	Bussmann	2	170M5408
271A-7	232	500	690	Bussmann	2	170M5410
330A-7	293	630	690	Bussmann	3	170M6410
370A-7	330	700	690	Bussmann	3	170M6411
430A-7	375	700	690	Bussmann	3	170M6411
						3AXD00000588487

Note:

- See also *Protecting the drive and input cable in short-circuits and against thermal overload (page 86)*.
- In multicable installations, install only one fuse per phase (not one fuse per conductor).
- Fuses with higher current rating than the recommended ones must not be used. Fuses with lower current rating can be used.
- Fuses from other manufacturers can be used if they agree with the ratings and the melting curve of the fuse does not exceed the melting curve of the fuse mentioned in the table.

Dimensions, weights and free space requirements

Standard drive module configuration drive module + LCL filter module)								
Frame size	**Height**		**Width**		**Depth**		**Weight***	
	mm	**in**	**mm**	**in**	**mm**	**in**	**kg**	**lb**
R11	1741	68.5	713	28.1	512	20.2	365	805

Optional selection +0B051+0H371 (without shrouds and full-size output power cable connection terminals)								
Frame size	**Height**		**Width**		**Depth**		**Weight***	
	mm	**in**	**mm**	**in**	**mm**	**in**	**kg**	**lb**
R11	1733	68.2	401	15.8	506	19.9	161	355

Optional selection +H381 (full power cabling panels)								
Frame size	**Height**		**Width**		**Depth**		**Weight***	
	mm	**in**	**mm**	**in**	**mm**	**in**	**kg**	**lb**
R11	1780	70.1	709	27.9	500	19.7	395	871

*approximate (depends on the selected options)

Weight of the LCL filter module: 180 kg (396 lb)

The weight of the cabling panels of option +H381 is 30 kg (66 lb)

Weight of optional selections										
Frame size	**+E208**		**+H356**		**+0H371**		**+H370**		**+0B051**	
	kg	**lb**	**kg**	**lb**	**kg**	**lb**	**kg**	**lb**	**kg**	**lb**
R11	3	7	2	4	-2.9	-6	2.9	6	-1.5	-3

For requirements of free space around the drive module, see *Required free space (page 58)*.

Losses, cooling data and noise

The air flow direction is from bottom to top.

This table shows typical heat loss values, required air flow and noise at the nominal ratings of the drive. The heat loss values can vary depending on voltage, cable conditions, motor efficiency and power factor. To obtain more accurate values for given conditions, use ABB DriveSize tool (http://new.abb.com/drives/software-tools/drivesize).

Drive type ACS880-34-	Frame size	Air flow		Heat dissipation	Noise
		m^3/h	ft^3/min	W	dB(A)
U_N = 400 V					
246A-3	R11	2100	1279	5280	75
293A-3	R11	2100	1279	6400	75
363A-3	R11	2100	1279	8000	75
442A-3	R11	2100	1279	10000	75
505A-3	R11	2100	1279	10000	75
585A-3	R11	2100	1279	12600	75
650A-3	R11	2100	1279	14200	75
U_N = 500 V					
240A-5	R11	2100	1279	5280	75
260A-5	R11	2100	1279	6400	75
302A-5	R11	2100	1279	8000	75
361A-5	R11	2100	1279	8000	75
414A-5	R11	2100	1279	10000	75
460A-5	R11	2100	1279	12600	75
503A-5	R11	2100	1279	14200	75
U_N = 690 V					
142A-7	R11	2100	1279	5280	75
174A-7	R11	2100	1279	6400	75
210A-7	R11	2100	1279	8000	75
271A-7	R11	2100	1279	10000	75
330A-7	R11	2100	1279	12600	75
370A-7	R11	2100	1279	14200	75
430A-7	R11	2100	1279	16000	75
					3AXD00000588487

The cooling air temperature rises 30 degrees Celsius when it goes through the drive module if the temperature of the input cooling air is 40 degrees Celsius.

Typical power cable sizes

The table below gives copper and aluminum cable types with concentric copper shield for the drives with nominal current. See also section *Terminal and entry data for the power cables (page 187)*.

Drive type ACS880-34-	IEC[1]		US [2]	
	Cu cable type	Al cable type	Cu cable type	Al cable type
	mm²	mm²	AWG/kcmil	AWG/kcmil
U_N = 400 V				
246A-3	2 × (3 × 50)	2 × (3 × 95)	250 MCM or 2 × 1	350 MCM or 2 × 2/0
293A-3	2 × (3 × 70)	2 × (3 × 120)	350 MCM or 2 × 2/0	500 MCM or 2 × 3/0
363A-3	3 × (3 × 50)	3 × (3 × 95)	500 MCM or 2 × 3/0	700 MCM or 2 × 4/0 or 3 × 2/0
442A-3	3 × (3 × 70)	3 × (3 × 120)	700 MCM or 2 × 4/0 or 3 × 2/0	2 × 300 MCM or 3 × 3/0
505A-3	3 × (3 × 70)	3 × (3 × 120)	700 MCM or 2 × 4/0 or 3 × 2/0	2 × 300 MCM or 3 × 3/0
585A-3	3 × (3 × 120)	3 × (3 × 150)	2 × 350 MCM or 3 × 4/0	2 × 500 MCM or 3 × 250 MCM
650A-3	3 × (3 × 120)	3 × (3 × 185)	2 × 400 MCM or 3 × 4/0	2 × 600 MCM or 3 × 300 MCM
U_N = 500 V				
240A-5	1 × (3 × 120)	2 × (3 × 70)	3/0	250 MCM or 2 × 1
260A-5	2 × (3 × 50)	3 × (3 × 70)	250 MCM or 2 × 1	350 MCM or 2 × 2/0
302A-5	2 × (3 × 70)	3 × (3 × 70)	300 MCM or 2 × 1/0	500 MCM or 2 × 3/0
361A-5	2 × (3 × 120)	3 × (3 × 70)	350 MCM or 2 × 2/0	500 MCM or 2 × 3/0
414A-5	3 × (3 × 50)	2 × (3 × 150)	500 MCM or 2 × 3/0	700 MCM or 2 × 4/0 or 3 × 2/0
460A-5	3 × (3 × 70)	3 × (3 × 120)	700 MCM or 2 × 4/0 or 3 × 2/0	2 × 300 MCM or 3 × 3/0
503A-5	3 × (3 × 95)	3 × (3 × 120)	2 × 250 MCM or 3 × 2/0	2 × 400 MCM or 3 × 4/0
U_N = 690 V				
142A-7	1 × (3 × 70)	2 × (3 × 50)	1/0	3/0
174A-7	2 × (3 × 50)	2 × (3 × 50)	2/0	4/0
210A-7	2 × (3 × 50)	2 × (3 × 70)	4/0	300 MCM or 2 × 1/0
271A-7	2 × (3 × 70)	3 × (3 × 50)	300 MCM or 2 × 1/0	400 MCM or 2 × 2/0
330A-7	2 × (3 × 50)	2 × (3 × 120)	400 MCM or 2 × 2/0	600 MCM or 2 × 4/0 or 3 × 1/0
370A-7	3 × (3 × 70)	2 × (3 × 150)	500 MCM or 2 × 3/0	700 MCM or 2 × 4/0 or 3 × 2/0
430A-7	3 × (3 × 70)	2 × (3 × 185)	700 MCM or 2 × 4/0 or 3 × 2/0	2 × 300 MCM or 3 × 3/0
				3AXD00000588487

[1] The cable sizing is based on max. 9 cables laid on a cable ladder side by side, three ladder type trays one on top of the other, ambient temperature 30 °C (86 °F) PVC insulation, surface temperature 70 °C (158 °F) (EN 60204-1 and IEC 60364-5-52). For other conditions, dimension the cables according to local safety regulations, appropriate input voltage and the load current of the drive.

[2] The cable sizing is based on NEC Table 310-16 for copper wires, 75 °C (167 °F) wire insulation at 40 °C (104 °F) ambient temperature. Not more than three current-carrying conductors in raceway or cable or earth (directly buried). For other conditions, dimension the cables according to local safety regulations, appropriate input voltage and the load current of the drive.

Terminal and entry data for the power cables

The maximum accepted cable size is 4 × (3 × 240) mm² or 4 × (3 × 500 AWG). Screw size for connecting busbars to the drive module input and output busbars: M12, tightening torque 50…75 N·m.

Units with optional cabling panels (+H381)

The maximum accepted cable size is 4 × (3 × 240) mm^2 or 4 × (3 × 500 AWG). The cabling panels are connected to the drive module busbars with M12 serpress nuts, tightening torque 30 N·m (20 lbf·ft).

Input, motor and brake resistor cable terminal sizes and tightening torques are given below.

L1/U1, L2/V1, L3/W1, T1/U2, T2/V2, T3/W2, UDC+, UDC-				Grounding busbar			
Screw		Tightening torque		Screw		Tightening torque	
		N·m	lbf·ft			N·m	lbf·ft
M12	1/2	50…75	37…55	M10	3/8	30…44	22…32

Two-hole 1/2 inch diameter cable lugs can be used.

Units without full-size output cable connection terminals (+0H371) and with a common mode filter (+E208)

It is possible to use the maximum cable size (4 × [3 × 240] mm^2 or 4 ×[(3 × 500 AWG]) only with special cable lugs and additional insulation. For more information, contact your local ABB representative.

Terminal data for the control cables

See *Connector data*.

Electrical power network specification

Voltage (U_1)	ACS880-34-xxxx-3 drive modules: 380…415 VAC 3-phase +10%…-15%. This is indicated in the type designation label as typical input voltage level 3~400 V AC. ACS880-34-xxxx-5 drive modules: 380…500 V AC 3-phase +10%…-15%. This is indicated in the type designation label as typical input voltage levels 3~400/480/500 V AC. ACS880-34-xxxx-7 drive modules: 525…690 V AC 3-phase +10%…-15%. This is indicated in the type designation label as typical input voltage levels 3~525/600/690 V AC (600 V AC UL, CSA).
Network type	TN (grounded) and IT (ungrounded) systems
Short-circuit withstand strength (IEC 61439-1)	Maximum allowable prospective short-circuit current is 65 kA when protected by the fuses given in the fuse table. For the maximum allowable prospective short-circuit current with circuit breakers, see section *Protecting the drive and the input power cable in short-circuits (page 86)*.
Short-circuit current protection (UL 61800-5-1)	The drive is suitable for use on a circuit capable of delivering not more than 100,000 rms symmetrical amperes at 600 V maximum when protected by the fuses given in the fuse table.
Short-circuit current protection (CSA C22.2 No. 14-05)	The drive is suitable for use on a circuit capable of delivering not more than 100 kA rms symmetrical amperes at 600 V maximum when protected by the fuses given in the fuse table.
Frequency (f_1)	50/60 Hz. Variation ±5% of nominal frequency.
Imbalance	Max. ± 3% of nominal phase to phase input voltage
Fundamental power factor (cos phi$_1$)	0.98 (at nominal load)

Harmonic distortion

Harmonics are below the limits defined in IEEE 519-2014, and G5/4. The drive complies with IEC 61000-3-2, IEC 61000-3-4 and IEC 61000-3-12.

The table below shows typical values of the drive for short-circuit ratio (I_{sc}/I_1) of 20 to 100. The values will be met if the supply network voltage is not distorted by other loads and when the drive operates at nominal load.

Nominal bus voltage V at PCC	THDi (%)	THDv (%)
V ≤ 690 V	3*	< 3**

THDv Indicates the total magnitude of the voltage distortion. This value is defined as the ratio (in %) of the harmonic voltage to the fundamental (non-harmonic) voltage:

THDi Indicates the total harmonic current distortion of the wave form. This value is defined as the ratio (in %) of the harmonic current to the fundamental (non-harmonic) current measured at a load point at the particular moment when the measurement is taken:

$$\text{THDv} = \frac{\sqrt{\sum_{2}^{40} U_n^2}}{U_1} \cdot 100\%$$

PCC Point on a public power supply system, electrically nearest to a particular load, at which other loads are, or could be, connected. The PCC is a point located upstream of the considered installation.

$$\text{THDi} = \frac{\sqrt{\sum_{2}^{40} I_n^2}}{I_1} \cdot 100\%$$

I_{sc}/I_1 Short-circuit ratio

I_{sc} Maximum short-circuit current at PCC

I_1 Continuous rms input current of the drive

I_n Amplitude of the current harmonic n

U_1 Supply voltage

U_n Amplitude of the voltage harmonic n

* The short-circuit ratio can influence the THDi value

** Other loads can influence the THDv value

Motor connection data

Motor types	Asynchronous AC induction motors, permanent magnet motors, AC induction servomotors and ABB synchronous reluctance motors (SynRM motors)
Voltage (U_2)	0 to U_1, 3-phase symmetrical. This is indicated in the type designation label as typical output voltage level as 3 0…U_1, U_{max} at the field weakening point.
Frequency (f_2)	0…500 Hz **Note:** Operation above 150 Hz can require type-specific derating. For more information, contact your local ABB representative. For drives with d*u*/d*t* filter: 120 Hz For drives with sine filter: 120 Hz
Frequency resolution	0.01 Hz

Current	See section *Ratings*.
Switching frequency	3 kHz (typically)
Maximum recommended motor cable length	DTC control: 500 m (1640 ft) Scalar control: 500 m (1640 ft) **Note:** For restrictions due to EMC compatibility, see section *EMC compliance (IEC/EN 61800-3:2004) (page 194)*. Longer motor cables cause a motor voltage decrease which limits the available motor power. The decrease depends on the motor cable length and characteristics. Contact ABB for more information. Note that a sine filter (optional) at the drive output also causes a voltage decrease.

DC connection data

Drive type ACS880-34-	Capacitance (mF)
U_N = 400 V	
246A-3	10.5
293A-3	10.5
363A-3	10.5
442A-3	10.5
505A-3	10.5
585A-3	14.0
650A-3	14.0
U_N = 500 V	
240A-5	10.5
260A-5	10.5
302A-5	10.5
361A-5	10.5
414A-5	10.5
460A-5	14.0
503A-5	14.0
U_N = 690 V	
142A-7	5.3
174A-7	5.3
210A-7	5.3
271A-7	5.3
330A-7	5.3
370A-7	5.3
430A-7	5.3

Control panel type

ACS-AP-W assistant control panel